

Business Analysis Case Study:

RFID-Enabled Cart Running and Shelf Placement Process for Medication Canisters

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Scope Document: RFID-Enabled Cart Running and Shelf Placement Process for Medication Canisters

Background & Current State:

At mail-order pharmacies/fulfillment centers the cart running process requires staff to scan individual medication canisters by barcode and place them into the correct shelf/bin location. This is a tedious and labor-intensive process and creates a bottleneck when hundreds of canisters must be shelved. The proposed solution is to replace repeated barcode confirmation steps with RFID-enabled canisters and RFID readers at each shelf/bin position so the system can automatically detect and register what item was placed and where, reducing inefficiencies in the process.

Problem Statement:

The existing barcode-based process is inefficient for high-volume canister replenishment because each canister must be manually scanned during placement on shelves. This increases handling time, and introduces opportunities for missed scans, incorrect placements, and workflow delays.

Scope:

- Review of the current cart running workflow for medication canister replenishment.
- Design of a future-state RFID-enabled shelf placement process
- Use of RFID tags on medication canisters
- Use of RFID readers or sensor-enabled shelf/bin positions
- Automatic capture of:
 - Canister identity
 - Bin/shelf location
 - Time of placement
- Integration with pharmacy management software.

Out of Scope:

- Redesign of operations outside cart running
- Modifications to pharmacy management software

Business Requirements Document (BRD)

1. Cart runners must be able to place items onto shelves without having to scan each canister individually.

- 1.1. Middleware software must automatically recognize the canister identity and location when a canister is placed in a bin/shelf location.
- 1.2. Middleware software must alert the cart runner of successful recognition of canister identity and location upon placement.
- 1.3. RFID readers must identify the placement of a canister into its location, and the following information must be transmitted:
 - 1.3.1. Canister identity
 - 1.3.2. Location code
 - 1.3.3. Timestamp of placement

2. Pharmacy management software must be accurately updated through the middleware software.

- 2.1. Middleware software must transmit canister identity and location to the pharmacy management software.
- 2.2. Pharmacy management software must update inventory records in near real-time upon receiving placement data.
- 2.3. The system must maintain an audit trail of all canister placement and movement events.

3. Cart runners must be able to place multiple canisters efficiently without delays.

- 3.1. Middleware software must support simultaneous detection of multiple RFID-tagged canisters.
- 3.2. The system must process high volumes of placement events without performance degradation or system failure.
- 3.3. RFID readers must reliably detect multiple canisters within defined shelf/bin zones.

4. The system must validate placement accuracy and handle exceptions.

- 4.1. Middleware software must validate that the canister is placed in the correct bin/shelf location.
- 4.2. The system must alert the cart runner when a canister is placed in an incorrect location.
- 4.3. The system must detect and flag unread RFID tags or failed reads.
- 4.4. The system must handle duplicate or conflicting reads and prevent incorrect inventory updates.

5. Cart runners must be able to manually scan canisters during RFID system downtime.

- 5.1. Shelves must have scannable QR/barcode labels in each bin/shelf location.
- 5.2. Middleware and pharmacy management software must support manual barcode-based placement as a fallback process.
- 5.3. The system must ensure continuity of operations during RFID outages without data loss.

6. The system must support usability and adoption by cart runners.

- 6.1. The placement process must be simple and require minimal user interaction.
- 6.2. The system must provide clear visual or audible feedback for successful and failed placements.
- 6.3. Training and standard operating procedures must support transition from barcode to RFID workflow.

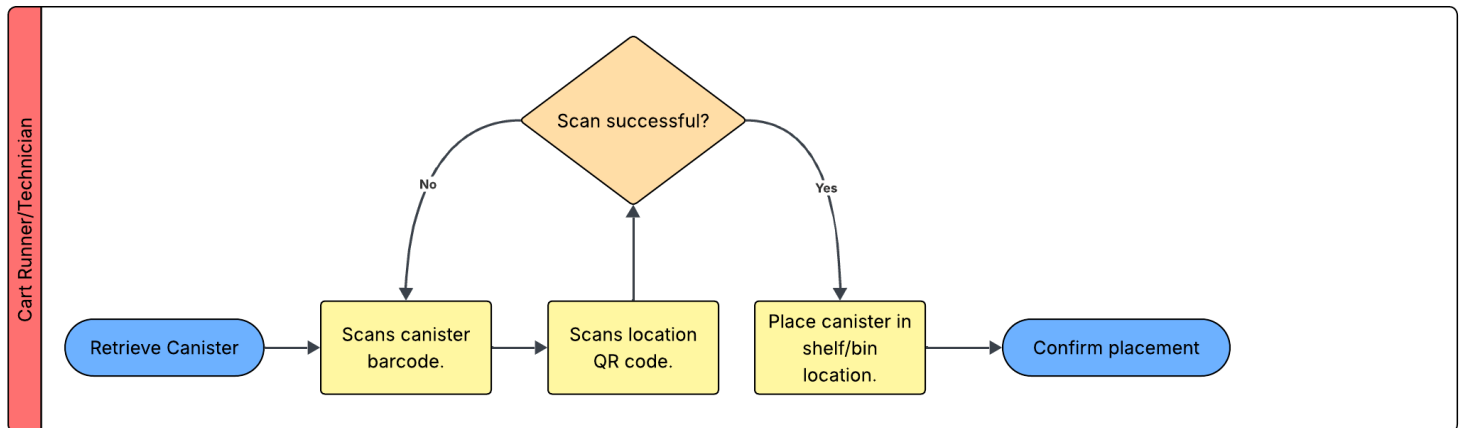
GAP Analysis

Theme	Current State	Future State	Gap / Roadblock	Required Changes (Action Plan)
1. Process Automation & Technology Enablement	Manual, barcode-driven workflow requiring scan of each canister and location; minimal shelf-level technology	RFID-enabled “place-and-detect” workflow with automated identification and bin-level tracking	Heavy reliance on manual scanning and lack of supporting infrastructure creates bottlenecks and limits scalability	1) Redesign workflow to eliminate scan steps 2) Tag canisters with RFID 3) Deploy RFID readers and antennas at shelf/bin level 4) Configure controlled read zones for accurate location detection 5) Implement RFID middleware for event processing 6) Run pilot to validate performance and efficiency gains

2. Data Visibility & Controls	Data captured only during manual scans; limited real-time visibility; accuracy depends on user actions	Real-time, automated capture of canister ID and location with system-driven validation and alerts	Lack of real-time data and reliance on manual input increases risk of errors and delayed inventory updates	1) Define data model (tag ID, location ID, event type) 2) Enable event-driven data capture from RFID reads 3) Configure business rules for placement validation 4) Implement real-time alerts for incorrect placement or unread tags 5) Integrate RFID data with inventory system 6) Establish audit trail for all placement events
3. Adoption & Operational Readiness	Staff follow repetitive scanning workflow; process is familiar but inefficient	Simplified, low-touch workflow with exception-based handling	Resistance to change and need for training on new process and technology	1) Develop training materials and updated SOPs 2) Conduct pilot with end-user participation 3) Gather feedback and refine workflow 4) Provide clear user guidance for exceptions and alerts 5) Maintain barcode fallback process for system exceptions 6) Roll out in phases with ongoing support

Activity Diagram

Current: Medication Canister Stocking Process



Future: Medication Canister Stocking Process

